Multistage amplifiers

Answers and worked solutions · Day 14 · Fri 6 Nov · pack E8

5. Practice

From your course files

1. Prac 4 calculation section

DC is unchanged from Worked Example 2 (capacitors are open at DC): V_B = 4.772 V, I_E = 4.072 mA, r_e = 26 mV/4.072 mA = 6.39 Ω in both stages, V_{CE} = 6.97 V.

Stage 2 is now bypassed, so it presents only βr_e to stage 1:

$$Z_{i_2} = 15\ \text{k} \parallel 4.7\ \text{k} \parallel 200(6.39) = 941\ \Omega$$

$$A_{v_1} = -\frac{R_C \parallel Z_{i_2}}{r_e} = -\frac{2.2\ \text{k} \parallel 941}{6.39} = -103.2, \qquad A_{v_{2(NL)}} = -\frac{R_C}{r_e} = -\frac{2200}{6.39} = -344.5$$

$$A_{v_{T(NL)}} = (-103.2)(-344.5) \approx \mathbf{35.6 \times 10^3}$$

This is the Week 5 lecture example (Worked Example 1), which gets 36.1×10^3 with the rounded r_e = 6.34 Ω.

Why step 7 sees clipping. The output can swing only about $V_{CE} - 0.2 \approx 6.8$ V before stage 2 saturates, so the input must stay below $6.8/35.6 \times 10^3 \approx 0.19$ mV peak. Even the handout's suggested 4 mV would ask for about 140 V at the output, so the output clips hard until V_i is well under 1 mV.

WS6 Q10 (course question).

Q_2's emitter is grounded, so its input resistance $r_{\pi 2}$ sits directly from Q_1's collector to ground, in parallel with R_C. Q_2's collector is at V_{CC} and does not affect v_{out}.

$$v_o = -g_{m1} v_{\pi 1}(R_C \parallel r_{\pi 2}), \qquad v_{in} = [r_{\pi 1} + (1+\beta)R_E]\frac{v_{\pi 1}}{r_{\pi 1}}$$

$$\frac{v_o}{v_{in}} = \frac{-g_{m1} r_{\pi 1}(R_C \parallel r_{\pi 2})}{r_{\pi 1} + (1+\beta)R_E} = \frac{-(R_C \parallel r_{\pi 2})}{\dfrac{1}{g_{m1}} + \dfrac{1+\beta}{\beta}R_E}$$

β here is β_1. The point for multistage work: stage 2's input impedance ($r_{\pi 2}$) loads stage 1's collector.

WS6 Q12 (course question).

I_X = 1 mA and I_Y = 0.5 mA are the collector currents, and both BJTs are active:

$$I_{B1} = \frac{I_X}{\beta} = 0.01\ \text{mA}, \qquad I_{B2} = \frac{I_Y}{\beta} = 0.005\ \text{mA}$$

KCL at node A (the joined bases): $I_1 = I_{B1} + I_{B2}$ = 0.015 mA. Both emitters are grounded, so $V_B = V_{BE} = 0.7$ V, and

$$V_{BB} = V_B + R_1 I_1 = 0.7 + 5(0.015) = \mathbf{0.775\ V}$$

WS6 Q13 (course question).

1. Start from Q_2, where the emitter current is known:

$$I_{C_2} = \alpha_2 I_{E_2} = 0.98(120) = \mathbf{117.6\ mA}, \qquad I_{B_2} = I_{E_1} = (1-\alpha_2) I_{E_2} = \mathbf{2.4\ mA}$$

Then Q_1:

$$I_{B_1} = (1-\alpha_1) I_{E_1} = 0.01(2.4) = \mathbf{0.024\ mA}, \qquad I_{C_1} = \alpha_1 I_{E_1} = \mathbf{2.376\ mA}$$

KCL at node A (the joined collectors): $I_C = I_{C_1} + I_{C_2}$ = **119.976 mA**.

2. KVL from 20 V through the 0.1 kΩ resistor:

$$v_o = 20 - 0.1 I_C = 20 - 0.1(119.976) = \mathbf{8.00\ V}$$

WS6 Q14 (course question).

READING THE FIGURE: $+20$ V and -10 V rails. Node A is Q_1's base, tied to ground through 10 kΩ. Q_1 is npn: collector C goes through 10 kΩ to +20 V, and emitter B goes through 9.3 kΩ to −10 V. Q_2 is pnp: its base is at node C, emitter D goes through 9.3 kΩ to +20 V, and collector E goes through 10 kΩ to −10 V.

1. $\beta = \infty$ means $I_{B1} = 0$. No current flows in the 10 kΩ resistor, so $V_A = \mathbf{0}$ V.
2. Q_1 is npn, so $V_B = V_A - 0.7 = \mathbf{-0.7}$ V.
3. $I_{E1} = I_{C1} = \dfrac{-0.7 - (-10)}{9.3\ \text{k}} = 1.00$ mA, so $V_C = 20 - (10\ \text{k})(1\ \text{mA}) = \mathbf{10.0}$ V.
   Q_1 check: $V_{CE1} = 10.7$ V > 0.2 V, so it is active ✓.
4. Q_2 is **pnp**, so its emitter sits 0.7 V *above* its base: $V_D = V_C + 0.7 = \mathbf{10.7}$ V.
5. $I_{E2} = I_{C2} = \dfrac{20 - 10.7}{9.3\ \text{k}} = 1.00$ mA, so $V_E = -10 + (10\ \text{k})(1\ \text{mA}) = \mathbf{0}$ V.
6. Q_2 check: $V_{EC2} = 10.7 - 0 = 10.7$ V > 0.2 V, so it is **active** ✓.

Where the official solution goes wrong: it writes $V_D = 0.7$ V, which treats the emitter as 0.7 V above *ground* instead of above the base (node C = 10 V). It then gets $I_{C2} = 2.075$ mA and $V_E = 10.75$ V, and concludes that Q_2 is saturated. That conclusion is wrong. $I_{B2} = 0$ does not disconnect the base from node C; it only means no current flows in that wire. The

> symmetric resistor values (10 k / 9.3 k on both sides) are a strong hint that the designer intended 1 mA in each transistor and both active.

WS6 Q16 (course question).

> **Stage 1** (exact Thevenin; C_{E_1} is open at DC, so the whole 2.2 kΩ counts):
>
> $$R_{Th} = 33\ \text{k} \parallel 10\ \text{k} = 7.67\ \text{k}\Omega, \qquad E_{Th} = \frac{10(14)}{43} = 3.26\ \text{V}$$
>
> $$I_{B_1} = \frac{3.26 - 0.7}{7.67 + 101(2.2)} = 11.1\ \mu\text{A}, \qquad I_{C_1} = 100 I_{B_1} = 1.11\ \text{mA}$$
>
> V_{E_1} = 2.47 V.
>
> **Shared node** ($V_{B_2} = V_{C_1}$, with I_{B_2} neglected, as the course does):
>
> $V_{B_2} = 14 - 1.11(6.8) = 6.44\ \text{V}, \qquad V_{E_2} = 6.44 - 0.7 = 5.74\ \text{V}, \qquad I_{E_2} = \frac{5.74}{1.2\ \text{k}} = 4.78\ \text{mA}$
>
> $V_{C_2} = V_{CC}$ = 14 V, so V_{CE_2} = 14 − 5.74 = **8.26 V**, and V_{CE_1} = 6.44 − 2.47 = 3.97 V. Both active.
>
> The official solution gives 11.17 µA, 1.12 mA, 6.38 V, 5.68 V, 4.73 mA and 8.32 V. Its denominator 229.2 kΩ should be 229.9 kΩ, and it rounds I_{C_1} up; the method is the same.
>
> > **NOTE ON I_{B_2}:** $\beta_2 = 50$, so $I_{B_2} = 4.78\ \text{mA}/51 = 94$ µA. That is about 8 % of I_{C_1}, and it flows through the 6.8 kΩ resistor. Including it with node KCL gives $V_{B_2} = 5.87$ V and $I_{E_2} = 4.30$ mA. The course answer neglects it. Give the course answer, then add one line saying it is an approximation.

WS6 Q17 (course question).

> Q_1 is a pnp whose collector drives the base of the npn Q_2; Q_1's emitter and Q_2's collector share the node under the 220 Ω resistor.
>
> **(a)** Loop from 12 V through 220 Ω (carrying $\approx I_{C_2} = \beta_1\beta_2 I_{B_1}$), Q_1's emitter-base junction and the 1.8 MΩ to ground:
>
> $$I_{B_1} = \frac{V_{CC} - V_{EB_1}}{R_B + \beta_1\beta_2 R_C} = \frac{12 - 0.7}{1.8\ \text{M} + 80(160)(220)} = \mathbf{2.45\ \mu A}$$
>
> $I_{C_1} = \beta_1 I_{B_1} = 80(2.45\ \mu\text{A}) = \mathbf{196\ \mu A} = I_{B_2}, \qquad I_{C_2} = \beta_2 I_{B_2} = 160(196\ \mu\text{A}) = \mathbf{31.36\ mA}$
>
> **(b)**
>
> - $V_{B_1} = I_{B_1} R_B = 2.45\ \mu\text{A} \times 1.8\ \text{M}\Omega$ = **4.41 V**

- $V_{E_1} = 12 - I_C(220\ \Omega) = 12 - 31.36(0.22)$ = **5.1 V** (check: pnp, so $V_{E_1} = V_{B_1} + 0.7$ = 5.11 V ✓)
- $V_{B_2} = V_{BE_2}$ = **0.7 V**, and $V_{C_1} = V_{B_2}$ = **0.7 V**
- $V_{C_2} = V_{E_1}$ = **5.1 V**, and V_{E_2} = **0 V**

The solution uses $I_C \approx I_{C_2}$ in the 220 Ω. Adding I_{E_1} (0.2 mA) would lower V_{E_1} to 5.06 V, which does not change the answer at this precision.

WS6 Q18 (course question).

1. The two BJTs are identical, have the same V_{BE} (bases joined, emitters joined), so they carry the same currents: $I_{B_1} = I_{B_2} = I_B$ and $I_{C_1} = I_{C_2} = I_o = \beta I_B$.

The base wire takes $I_x = I_{B_1} + I_{B_2} = 2I_B$ from Q_1's collector node. KCL at that node:

$$I_{REF} = I_{C_1} + I_x = I_o + 2I_B = I_o + \frac{2I_o}{\beta} = I_o\left(1 + \frac{2}{\beta}\right)$$

$$\boxed{\frac{I_o}{I_{REF}} = \frac{1}{1 + \dfrac{2}{\beta}}}$$

2. For very large β, $2/\beta \approx 0$, so $I_o/I_{REF} = 1$: the output current copies the reference. (With β = 100 the error is 2 %.)

WS6 Q19 (course question).

1. Assume active. The current source sets I_E = 1 mA, so

$$I_B = \frac{I_E}{1+\beta} = \frac{1}{101} = 0.0099\ \text{mA}, \qquad I_C = I_E - I_B = 0.9901\ \text{mA}$$

The base current flows out of ground through R_B into the base:

- $V_B = -R_B I_B = -100(0.0099)$ = **−0.990 V**
- $V_E = V_B - V_{BE} = -0.990 - 0.7$ = **−1.690 V**
- $V_C = V_{CC} - R_C I_C = 10 - 7.5(0.9901)$ = **2.574 V**

Check: $I_B > 0$ and $V_{CE} = 2.574 - (-1.690) = 4.26$ V > 0.2 V, so active.

2. With very large β, $I_{REF} = I$ = 1 mA. Q_1 is diode-connected, so $V_{C_1} = V_{B_1} = -V_{EE} + 0.7 = -10 + 0.7 = -9.3$ V. The resistor runs from V_{CC} to Q_1's collector:

$$R = \frac{V_{CC} - V_{C_1}}{I_{REF}} = \frac{10 - (-9.3)}{1\ \text{mA}} = \mathbf{19.3\ k\Omega}$$

New questions

N1 [5 marks]. Two-port cascade.

a. Stage 1 drives Z_{i_2} through Z_{o_1}:

$$A_{v_1} = A_{v_{NL1}} \frac{Z_{i_2}}{Z_{i_2} + Z_{o_1}} = -120 \cdot \frac{1.5}{4.5} = -40$$

Stage 2 drives R_L through Z_{o_2}:

$$A_{v_2} = -80 \cdot \frac{6}{10} = -48$$

$$A_{v_T} = (-40)(-48) = \mathbf{1920}$$

b. $A_{v_s} = \dfrac{2}{2+1}(1920) = \mathbf{1280}$

c. $A_{v_{NL1}} A_{v_{NL2}} = 9600$, so the loading costs a factor of $9600/1280 = \mathbf{7.5}$. Most of the loss is at the interstage divider (×1/3), because $Z_{i_2} < Z_{o_1}$.

N2 [10 marks]. Two-stage CE–CE cascade.

a. $\beta R_E = 150\ \text{k}\Omega \geq 10 R_2 = 100\ \text{k}\Omega$ ✓, so the approximate method holds (just).

$$V_B = \frac{10}{57}(12) = 2.105\ \text{V}, \quad V_E = 1.405\ \text{V}, \quad I_E = 1.405\ \text{mA}$$

$$V_{CE} = 12 - 1.405(3.3 + 1) = 5.96\ \text{V (active)}, \quad r_e = \frac{26}{1.405} = 18.5\ \Omega$$

The two stages are identical. (The exact Thévenin method gives $I_E = 1.33$ mA. That is a 5 % difference because the 10× rule is only just met.)

b. $Z_{i_2} = 47\ \text{k} \parallel 10\ \text{k} \parallel (150)(18.5) = 47\ \text{k} \parallel 10\ \text{k} \parallel 2.775\ \text{k} = \mathbf{2.08\ k\Omega}$

$$A_{v_1} = -\frac{3.3\ \text{k} \parallel 2.08\ \text{k}}{18.5} = -\frac{1274}{18.5} = \mathbf{-68.9}$$

c. $A_{v_2} = -\dfrac{3.3\ \text{k} \parallel 10\ \text{k}}{18.5} = -\dfrac{2481}{18.5} = \mathbf{-134.1}$

$$A_{v_T} = (-68.9)(-134.1) = \mathbf{+9.24 \times 10^3}$$

d. $Z_{i_1} = Z_{i_2} = \mathbf{2.08\ k\Omega}$ (identical stage).

$$A_{v_s} = \frac{2.08}{2.08 + 0.6}(9238) = \mathbf{7.17 \times 10^3}$$

$$v_o = (7167)(20\ \mu\text{V}) = \mathbf{143}\ \text{mV peak}$$

This is in phase with v_s, and well inside the swing allowed by $V_{CE} \approx 6$ V.

N3 [10 marks]. CE–EF cascade (buffering a low-resistance load).

a. Checks: $\beta R_{E1} = 144\text{ k} \geq 120\text{ k}$ ✓ and $\beta R_{E2} = 264\text{ k} \geq 220\text{ k}$ ✓. Stage 1: $V_{B1} = \frac{12}{68}(15) = 2.647$ V, so $I_{E1} = \frac{1.947}{1.2\text{ k}} = \mathbf{1.62}\text{ mA}$ and $r_{e1} = \mathbf{16.0}\ \boldsymbol{\Omega}$. Also $V_{C1} = 7.37$ V and $V_{CE1} = 5.43$ V, so Q1 is active. Stage 2: $V_{B2} = 7.5$ V and $V_{E2} = 6.8$ V, so $I_{E2} = \frac{6.8}{2.2\text{ k}} = \mathbf{3.09}\text{ mA}$ and $r_{e2} = \mathbf{8.41}\ \boldsymbol{\Omega}$. $V_{CE2} = 8.2$ V, so Q2 is active.

b. The follower's emitter load is $R_{E2} \parallel R_L = 2.2\text{ k} \parallel 500 = 407.4\ \Omega$.

$$Z_{b2} = \beta(r_{e2} + R_{E2} \parallel R_L) = 120(415.8) = 49.9\text{ k}\Omega$$

$$Z_{i_2} = 22\text{ k} \parallel 22\text{ k} \parallel 49.9\text{ k} = \mathbf{9.01}\ \mathbf{k\Omega}$$

c. $A_{v_1} = -\dfrac{4.7\text{ k} \parallel 9.01\text{ k}}{16.0} = -\dfrac{3089}{16.0} = \mathbf{-192.8}$

$$A_{v_2} = \frac{407.4}{8.41 + 407.4} = \mathbf{0.980}$$

$$A_{v_T} = \mathbf{-188.9}$$

d. $A_v = -\dfrac{4.7\text{ k} \parallel 500}{16.0} = \mathbf{-28.2}$

The follower raises the gain by a factor of about 6.7. A 500 Ω load directly on a 4.7 kΩ output impedance collapses the CE gain. The follower presents about 9 kΩ to Q1 instead. It also has a low output impedance, $Z_{o2} = R_{E2} \parallel \left(r_{e2} + \frac{R_C \| R_3 \| R_4}{\beta}\right) \approx 35\ \Omega$, so the 500 Ω load barely loads it.

N4 [7 marks]. Darlington emitter follower.

a. $\beta_D = (100)(80) = \mathbf{8000}$

$$I_B = \frac{18 - 1.6}{3.3\text{ M} + (8000)(390)} = \frac{16.4}{6.42\text{ M}} = \mathbf{2.55}\ \boldsymbol{\mu}\mathbf{A}$$

$I_E \approx \beta_D I_B = \mathbf{20.4}\text{ mA}$, $\quad V_E = (20.4\text{ mA})(390) = \mathbf{7.97}\text{ V}$, $\quad V_{CE} = 18 - 7.97 = \mathbf{10.0}\text{ V}$

b. $Z_{i_2} = \beta_D R_E = 3.12\ \text{M}\Omega$, so $Z_i = 3.3\text{ M} \parallel 3.12\text{ M} = \mathbf{1.60}\ \mathbf{M\Omega}$

$$A_i = \frac{\beta_D R_B}{R_B + \beta_D R_E} = \frac{8000(3.3)}{6.42} = \mathbf{4.11 \times 10^3}, \qquad A_v \approx \mathbf{1}\ (\text{in phase})$$

Note that R_B now limits Z_i. The Darlington itself presents 3.1 MΩ.

N5 [6 marks]. Current mirror design.

a. $V_{C1} = V_{B1} = 0.7$ V, so $R_{REF} = \dfrac{12 - 0.7}{2\ \text{mA}} = \mathbf{5.65\ k\Omega}$.

b. $I_{REF} = 2$ mA, so $I_o = \dfrac{I_{REF}}{1 + 2/\beta} = \dfrac{2}{1.025} = \mathbf{1.95\ mA}$ (2.4 % low).

c. $I_{REF} = I_o(1 + 2/\beta) = 2.05$ mA, so $R_{REF} = \dfrac{11.3}{2.05\ \text{mA}} = \mathbf{5.51\ k\Omega}$.

d. Q2 is active while $V_{CE2} \geq 0.2$ V: $R_L \leq \dfrac{12 - 0.2}{2\ \text{mA}} = \mathbf{5.9\ k\Omega}$.

6. Quick recall check

Question

1. Stage 1's gain changes, because its collector resistor is now in parallel with Z_{i_2}: $A_{v_1} = -(R_C \parallel Z_{i_2})/r_e$.
2. $Z_{i_2} = R_1 \parallel R_2 \parallel \beta(r_e + R_E)$.
3. $\beta_D = 6000$, $Z_i \approx \beta_D R_E = 6$ MΩ.
4. Q1 sees only r_{e2} as its load, so its gain is about −1. Its collector–base capacitance is therefore barely multiplied (little Miller effect), while Q2 (common base) supplies the full gain R_C/r_e.
5. $1/(1 + 2/50) = 0.962$.